The visible side

Mathematics before the session

There are three two-sided cards: RR has faces 1 and 2; RB has faces 3 and 4; BB has faces 5 and 6. Their face colors in order are R, R, R, B, B, B. A ticket selects one specific face to show. Under the six-ticket rule every ticket has probability 1/6, with replacement and fresh mixing between rounds.

Ticket	Showing color	Hidden color
1 or 2	R	R
3	R	B
4	B	R
5 or 6	B	B

If the only clue is red showing, the surviving equally likely tickets are 1, 2, 3. Two hide red and one hides blue, so P(hidden red | showing red) = 2/3. Likewise, given blue showing, tickets 4, 5, 6 remain and hidden blue has probability 2/3. The two possible card types after a red clue are not equally likely: RR supplied two eligible faces.

Changing the chooser changes the answer. With only tickets 1 and 3 selected uniformly, red always shows; one ticket hides red and one blue. Each hidden color then has probability 1/2. The visible clue is the same, but its production mechanism is different. This is the conditional-sample-space point of Bertrand’s box paradox.

Exact design condition

For a subset S of the six distinct tickets chosen uniformly, a red clue is fair precisely when ticket 3 is present and exactly one of tickets 1 and 2 is present. Any subset of {4,5,6} may also be present, since those tickets cannot give a red clue. Thus there are $2 \times 2^3 = 16$ fair subsets; only two are two-ticket cups. Keeping only whole cards can never achieve this condition, because faces 1 and 2 must then be kept or removed together. A cup that never shows red is excluded.

Assumptions and discoveries by band

The face number, card identity, unshown side, orientation process, and distinctive physical marks must stay hidden from the guesser. Revealing the number determines the hidden color exactly, and a partial fact filters the surviving faces: after a red clue, “odd” leaves faces 1 and 3 (a tie), “even” leaves face 2 (red for certain), and “3 or less” removes nothing (red stays 2 to 1). Equal-size tickets and cards, a screen, and identical covers make “color alone” an executable information rule. Repeated outcomes do not prove a conditional probability.

K-1: sort face possibilities, remove or add tickets, and compare a color clue’s surviving cases. **Grades 2-3:** pair tickets by showing and hidden color, judge both clues, and contrast the two chooser rules. **Grades 4-5:** diagnose unequal conditional weights, weigh partial information about the covered number, and characterize all fair subsets.

Running the session

Materials and preparation

Make three same-size, opaque, two-sided cards per kit: RR numbered 1/2, RB numbered 3/4, BB numbered 5/6. Number each face in the same corner. Add six same-size tickets numbered 1-6, a cup, a folded folder as screen, and identical opaque covers for the face numbers. For eleven children, five kits let every pair play the opening rounds (two pairs at the K-1 table, two at 2-3, and at 4-5 a pair with the third child as referee); three kits suffice if pairs take turns. Use R/B letters along with colors and check that no reverse-side printing shows through.

Print four pages per band. Readiness is matching two-sided objects, sorting by a clue, and small counts; fractions are optional. Adults may handle hidden orientation while children arrange the tickets as the proof. Give everyone time to inspect the known cards before hiding anything. For the card-removal problems, children lay each card’s two tickets on it, so removing a card takes both. Prepared, unpiloted material.

Short shared launch

Show the three cards face up. Show that ticket 4 picks the blue face of RB and its other face is red, as in the picture on page 1 of every packet. Then play one round behind the screen: draw a ticket, set its numbered face upward, keep the ticket hidden, cover the number, and show only the color. Return and mix the ticket after each round. Ask which tickets could have made the color showing, rather than asking children to guess the adult’s intention.

First rounds in pairs

At the tables, pairs play four to six rounds before Problem 1. The chooser works behind the screen as in the launch; the guesser names a hidden color, then flips the card to check; then they swap. Keep no tally: the rounds teach the game, and the problems replace long trials with the complete list of six tickets.

35-40 minutes at the tables

Launch: about 5 minutes, whole group. **Tables, 0-6 minutes:** paired rounds. **6-32:** the first route, Problems 1-3 in each band (page 9). K-1 sorts the six tickets and finds the tying removals for each clue; Grades 2-3 find the four pairs and the better guess for each clue; Grades 4-5 find the exact chances, answer the friend, and weigh the three facts about the covered number. **32-38:** check one claim with the tickets, or start Problem 4. **Sharing, a few minutes:** “After a red clue, which hidden color would you bet on, and which tickets tell you?” Later problems are return visits. A complete three-ticket argument is a good endpoint.

Each problem’s key, on pages 3-8, ends with a hint to use only if needed and an extension for a child who finishes.

Source and adaptation

Background: Brian Mintz, Dartmouth Graduate Student Seminar, Fall 2022, *Probability Paradoxes: Eleven Enlightening Examples*, slides, example 3, “Betrand's Box Paradox” [sic] (seminar page). The six-ticket protocol, subset designs and exact answers here are independently derived classroom adaptations.

K to 1 answer key

Problem 1

Hidden red: tickets **1, 2, 4**. Hidden blue: **3, 5, 6**. Same showing color but different hidden colors: 1 and 3, or 2 and 3, both show red; 4 and 5, or 4 and 6, both show blue. These four unordered pairs are all such choices.

Hint. Hold a ticket against the face it picks, then flip that card: which bin does the ticket go in? **Extension.** Which pairs of tickets show different colors but hide the same color? ({1,4}, {2,4}, {3,5}, {3,6}.)

Problem 2

Given red, hidden red has more ways: tickets 1 and 2 versus ticket 3. Removing **1 or 2** makes a tie, one surviving red-clue ticket for each hidden color. Removing 3 leaves only hidden red; removing 4, 5, or 6 changes no red-clue possibility.

Hint. Put only the tickets that show red in front of you. Which bin does each one go in? **Extension.** Which two tickets can you remove so that red showing always hides blue? (Only 1 and 2.)

Problem 3

Restore all tickets first. Given blue, hidden blue has two ways (5,6), hidden red one (4). Remove **5 or 6** to make a tie. Removing 4 leaves only hidden blue; removing 1, 2, or 3 changes no blue-clue possibility.

Hint. Which tickets show blue? **Extension.** Which two tickets can you remove so that blue showing always hides red? (Only 5 and 6.)

Problem 4

With {1,3}, neither hidden color has more ways: ticket 1 hides red and ticket 3 hides blue. **Given red showing**, adding **2** favors hidden red, two ways versus one. Adding **4, 5, or 6** leaves a tie because the added ticket shows blue and cannot have produced the red clue.

If a child instead counts hidden colors over *all* rounds, adding 2 or 4 favors red 2-to-1; adding 5 or 6 favors blue 2-to-1. Those counts are correct for that different question. Name the conditioning clue before comparing answers.

Each ticket still picks its own face, as on page 1, so an added 4, 5 or 6 shows blue. The two-ticket chooser always shows red, but an expanded cup may not; use the complete ticket set to explain why blue-showing additions cannot change a red-clue comparison.

Hint. Set aside the rounds where blue shows. Which tickets are left? **Extension.** Can any added tickets make blue the better guess after a red clue? (No: ticket 3, the only red-showing ticket that hides blue, is already in the cup, so additions can only add hidden red, with ticket 2, or nothing.)

K to 1 answer key, continued

Problem 5

The complete three-ticket choices are **{3,4,5}, {3,4,6}, {3,5,6}**. Ticket 3 must be included: it is the only red-showing face that hides blue. Tickets 1 and 2 must be excluded because they would show red and hide red. Choose the other two tickets from 4, 5, 6. Every listed cup sometimes shows red and then always hides blue. Two of the five answer groups stay empty.

Hint. Which red-showing ticket hides blue? **Extension.** Choose four tickets with the same rule. (Only {3,4,5,6}.)

Problem 6

Remove the **RB card**, meaning tickets 3 and 4, to make red showing always hide red. Remove the **RR card**, tickets 1 and 2, to make red showing always hide blue. In both cases a red clue remains possible. Removing BB leaves the original 2-to-1 red-clue weighting.

Hint. Keep the tickets on their cards. When a card goes, which red-showing tickets are left? **Extension.** Which card would you remove so that blue showing always hides red? (BB, leaving ticket 4 as the only blue face.)

Problem 7

Exactly **{1,3}** and **{2,3}**. A fair red clue needs one red-showing ticket hiding red (1 or 2) and one hiding blue (3). A blue-showing ticket uses up one of the two places without contributing to either red-clue category, so no other two-ticket cup works.

Hint. Start with ticket 3 in the cup. Which ticket could go with it? **Extension.** Find every three-ticket cup with a tie after a red clue. ({1,3} or {2,3} with one of 4, 5, 6: six cups.)

Grades 2 to 3 answer key

Problem 1

Ticket	Showing	Hidden
1	R	R
2	R	R
3	R	B
4	B	R
5	B	B
6	B	B

The four pairs are **{1,3}** and **{2,3}**, which show red, and **{4,5}** and **{4,6}**, which show blue. Two tickets that show the same color both come from 1, 2, 3 or both from 4, 5, 6. Among 1, 2, 3 only ticket 3 hides blue, so the pair is 3 with 1 or 2; among 4, 5, 6 only ticket 4 hides red, so the pair is 4 with 5 or 6. Two of the six answer slots stay empty.

Hint. Flip each card to see what each ticket hides, then compare tickets that show the same color. **Extension.** Which pairs show different colors but hide the same color? ({1,4}, {2,4}, {3,5}, {3,6}.)

Problem 2

Red is the better hidden-color guess. Red-showing tickets 1, 2, 3 are the surviving equally likely choices. Tickets 1 and 2 hide red; ticket 3 hides blue. Hence red has probability 2/3 and blue 1/3, given that red shows. Blue-showing tickets do not count in this conditional sample space.

Hint. Which tickets could have made red show? **Extension.** If you always guess that the hidden color matches the showing color, how often are you right over all six tickets? (Tickets 1, 2, 5, 6: four times in six.)

Problem 3

The better guess changes to blue. Blue-showing tickets 4, 5, 6 survive: 4 hides red; 5 and 6 hide blue. Hidden blue has probability 2/3. The rule “guess the showing color” wins two of three clue-compatible face outcomes for either clue.

Hint. Which tickets show blue? **Extension.** Which one-ticket removals make a blue clue a tie? (5 or 6.)

Grades 2 to 3 answer key, continued

Problem 4

They do not give the same hidden-color chances. Under the {1,3} chooser, red always shows and one of two equally likely tickets hides each color, so hidden red has chance 1/2. Under the six-ticket chooser restricted to red-showing rounds, tickets 1, 2, 3 remain, so hidden red has chance 2/3. The words “red is showing” are the same in both experiments; the rule that generated those words differs.

Hint. For each chooser, list the tickets that can give a red clue and what each hides. **Extension.** Add ticket 2 to the two-ticket chooser. Which rule does it now match? (Hidden red 2 of 3, as in the six-ticket rule, since 4, 5, 6 never give a red clue.)

Problem 5

Remove **RB** (tickets 3,4): red can still show and then certainly hides red. Remove **RR** (1,2): red can still show and then certainly hides blue. Remove **BB** (5,6): the red-clue odds stay 2-to-1 for red. None of the three whole-card removals gives equal hidden-color chances under a red clue.

Hint. Keep the tickets on their cards. Take one card away and list the red-showing tickets left. **Extension.** Remove two cards. When can red still show, and what does it hide? (Keep only RR: always red. Keep only RB: always blue. Keep only BB: red never shows.)

Problem 6

The only pairs are **{1,3}** and **{2,3}**. Ticket 3 is the sole red-showing/blue-hidden outcome. To balance it, choose exactly one of 1 or 2. If the pair contains a blue-showing ticket instead, only one red-showing ticket remains and its hidden color is certain; if neither shows red, the required clue is impossible.

Hint. A tie needs one red-showing ticket that hides red and one that hides blue. **Extension.** Allow three tickets. Find every cup with a tie after a red clue. ({1,3} or {2,3} with one of 4, 5, 6: six cups.)

Grades 4 to 5 answer key

Problem 1

Given red showing, tickets 1,2,3 remain equally likely: hidden red 2/3, hidden blue 1/3. Given blue showing, tickets 4,5,6 remain: hidden red 1/3, hidden blue 2/3. Count face tickets rather than merely the two card types compatible with the clue.

Hint. List the six equally likely tickets and keep those that show red. **Extension.** Before any clue, what is the chance that the hidden color matches the showing color? (Tickets 1, 2, 5, 6: 4/6 = 2/3.)

Problem 2

The argument is unsupported. Before conditioning, each card has two of the six face tickets. But both RR tickets show red, whereas only one RB ticket does. After retaining red-showing rounds, RR therefore has twice RB’s weight. The three surviving face outcomes, not the two surviving card names, are equiprobable.

Hint. How many red-showing tickets does each card supply? **Extension.** Remove one ticket so the friend’s conclusion becomes true. (Remove 1 or 2: then RR and RB each supply one red-showing ticket.)

Problem 3

After a red clue the equally likely faces are 1, 2, 3, hiding red, red, blue. **The number** decides: 1 or 2 means red, 3 means blue. **Odd** leaves faces 1 and 3: a tie. **Even** leaves face 2: red for certain. **3 or less** is true of all three faces, so nothing changes and red stays the better guess, 2/3. Each fact keeps only the faces it allows; this is why the original game covers the number.

Hint. After a red clue, list the faces that could be showing; keep only those the fact allows. **Extension.** Can a fact about the number make blue the better guess without making it certain? (No: face 3 is the only red face hiding blue, so any fact that favors blue leaves face 3 alone.)

Problem 4

Yes. With tickets 1 and 3, red always shows and the hidden-red chance is 1/2. With all six tickets conditioned on red, it is 2/3. One protocol gives each red-capable card one eligible face ticket; the other gives the RR card two. Both protocols are fully specified, so no assumption about a host’s unstated preferences is needed.

Hint. In each rule, which equally likely tickets can give the red clue? **Extension.** Which cups of distinct tickets give hidden red exactly 2/3 after a red clue? ({1,2,3} with any subset of {4,5,6}: eight cups.)

Grades 4 to 5 answer key, continued

Problem 5

No. Six hidden reds can occur under both rules, with conditional probabilities $(2/3)^6$ and $(1/2)^6$ respectively. Every six-color binary sequence is possible under both. The exact mechanisms prove the distributions differ; a short sample does not logically separate them.

Hint. Is any sequence of six hidden colors impossible under either rule? **Extension.** Compare how likely six hidden reds are under each rule. ($(2/3)^6 = 64/729$, about 0.088, against $(1/2)^6 = 1/64$, about 0.016: about 5.6 times as likely under the six-ticket rule, yet possible under both.)

Problem 6

Every solution has the form **{1,3} ∪ T** or **{2,3} ∪ T**, where T is any subset of {4,5,6}. Explicitly T may be ∅, {4}, {5}, {6}, {4,5}, {4,6}, {5,6}, or {4,5,6}. These two families give all 16 subsets, without repeats.

Proof of completeness: among red-showing tickets, hidden blue has only ticket 3 available, so it must be included. For equal positive weight, exactly one hidden-red ticket, 1 or 2, must be included. Blue-showing tickets are unrestricted because conditioning on red removes them.

Hint. Blue-showing tickets never give a red clue. Which red-showing tickets must be in the cup? **Extension.** Which subsets are fair for a blue clue? How many are fair for both clues? (Blue: ticket 4 and exactly one of 5, 6, with 1, 2, 3 free: 16. Both: 3 with one of 1, 2, and 4 with one of 5, 6: 4.)

Problem 7

No. Keeping RR supplies both hidden-red red-showing tickets or neither; keeping RB supplies the one hidden-blue red-showing ticket or none. A positive tie would require exactly one of 1,2 plus 3, which whole-card selection cannot do. BB supplies no red faces and cannot repair the imbalance. Exclude a BB-only selection because red never shows.

Hint. Which tickets come and go together when a card is kept or removed? **Extension.** Allow copies of whole cards. Which collections make a red clue fair, and which hide red 3 times in 4? (With a copies of RR and b of RB, hidden red has 2a ways and hidden blue b. Fair when $b = 2a$, such as one RR with two RB; red 3 times in 4 when $2a = 3b$, such as three RR with two RB. BB copies change nothing.)

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First routes, in 35-40 minutes at the tables after the launch and four to six paired rounds (page 2): K-1 P1-3 for the six-face possibilities and ticket removals; grades 2-3 P1-3 for the four pairs and both visible clues; grades 4-5 P1-3 for conditional weights and what a fact about the covered number reveals. On return, K-1 can add tickets or design a cup (P4-7). Grades 2-3 can compare chooser rules or find all fair two-ticket cups (P4-6). Grades 4-5 can compare the choosers (P4-5), then characterize every fair subset and the whole-card obstruction (P6-7). Grades 4-5 can then continue with the bonus companion, week-45-bonus.pdf with its own guide: repeated clues from the same card, two door hosts and a noisy reporter; grades 2-3 can try its first page. Keep the screen protocol unchanged so the mathematical clue remains color alone.